

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch : Artificial Intelligence & Allied

Semester : V

Subject Code & Name: BTAIPE504B – Soft Computing

Max Marks: 60

Date: 08-01-2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		(12)
A) Explain the anatomy of an Artificial Neural Network?	Apply	6
B) Differentiate between Soft Computing and Hard Computing techniques?	Remember	6
C) Explain Perceptron learning algorithm along with its types?	Synthesis	6
Q.2 Solve Any Two of the following.		(12)
A) Define Gradient Descent? Explain working of Gradient Descent?	Synthesis	6
B) What is an Activation function and why do you use them?	Understand	6
C) Explain any two Loss functions in ANN?	Remember	6
Q.3 Solve Any Two of the following.		(12)
A) Explain the structure of Multi-layer Perceptron with a neat diagram?	Understand	6
B) Draw the Architecture of Back-Propagation network and write its training algorithm?	Remember	6
C) What is Sigmoid Neurons? Mention its importance?	Apply	6
Q.4 Solve Any Two of the following.		(12)
A) Explain several Tensor operations that can be performed on Artificial Neural Networks?	Remember	6
B) How will you develop a Neural Network using PyTorch with proper steps? Explain with an example?	Apply	6
C) Explain techniques to reduce Overfitting in Artificial Neural Networks?	Synthesis	6
Q.5 Solve Any Two of the following.		(12)
A) Define Fuzzy Logic? Draw & explain Architecture of Fuzzy Logic System?	Remember	6
B) Explain the difference between Classical Set Theory and Fuzzy Set Theory? Mention few applications of Fuzzy Logic?	Apply	6
C) Explain several operators of Genetic Algorithms?	Synthesis	6

***** End *****